GAURAV KUMAR

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Education

- **National Institute of Technology Raipur** — Raipur, Chhattisgarh
  Bachelor of Technology - Electronics and Communication Engineering; — *2022 - 2026*
  Courses: *Operating Systems, C++, Data Structure and Algorithm , Object Oriented Programming,Computer Networks*

Skills Summary

- **Programming Languages**: Rust, Solidity, TypeScript, JavaScript, C++, SQL, Noir
- **Blockchain & Web3**: Ethereum, Solana, Smart Contracts, Solana Programs, Anchor, OpenZeppelin, Chainlink CRE, EVM, SPL Tokens, PDAs, Solana Web3.js / Solana Kit, Wallet Integration
- **Zero-Knowledge & Cryptography**: ZKPs, zk-SNARKs, Noir, Poseidon Hashing
- **Backend & Databases**: Next.js, React.js, Node.js, Axum, Actix Web, REST APIs, PostgreSQL, MongoDB, MySQL
- **DevOps & Tools**: Docker, Kubernetes, AWS, Linux/Ubuntu, Git, GitHub, Postman, Vim

Software Development Experience

- **Nxtwave Disruptive Technologies Private Limited** — Sep 2026 – Present
  Software Developer & Instructor — *Hyderabad, India*
  - **Full-Stack Development**: Developed an internal **Next.js** Instructor Evaluation Platform serving **900+ students** and **60+ instructors**, implementing student progress tracking, interaction logging, and instructor workflows.
  - **Backend & Database**: Built backend functionality using **Next.js API Routes/Server Actions** with **PostgreSQL**, designing database schemas for student profiles, instructor interactions, progress tracking, and historical records.
  - **Authentication & Access Control**: Implemented **authentication, session management, and role-based access control (RBAC)**, including **4-hour session expiry** and instructor-specific permissions to restrict interaction logging to assigned students.
- **National Institute of Technology Patna – Summer Internship** — May 2025 – June 2025
  Research Intern — *Patna, India*
  - **Full-Stack Development**: Built a full-stack IoT smart agriculture platform using **React.js** and **REST APIs**, processing **1000+ sensor data points daily**.
  - **Authentication & Deployment**: Implemented **token-based authentication** using Clerk and integrated real-time IoT monitoring and automation workflows for remote device management.

Projects

- **Loomzk: Stablecoin Lending on Solana [Live Demo] [GitHub]**:
  Tools & Technologies: **Rust**, **Anchor**, **Solana**, **Circom**, **Groth16**, **TypeScript**, **Next.js**, **Pyth**, **Chainlink**, **LiteSVM**.
  Built **Rust/Anchor smart contracts** for SOL-collateralized USDC/USDT loans (2x collateral, fixed 12% rate) with **Pyth and Chainlink oracle** validation and rule-based liquidation.
  Implemented **zero-knowledge repayment certificates** (Circom, Groth16) verified on-chain, with 281 automated tests in LiteSVM and a Next.js Proof of Reserves page. *Oct 2026*

- **Solana Prediction Market [Live Demo] [GitHub]**:
  Tools & Technologies: **Solana**, **Rust**, **Anchor**, **Next.js**, **React**, **TypeScript**, **Tailwind CSS**, **Codama**, **PDA**, **LiteSVM**.
  Built a full-stack **decentralized prediction market dApp** on Solana supporting YES/NO markets, SOL betting, outcome resolution, and proportional payouts.
  Engineered **Rust/Anchor smart contracts** using **PDA-based accounts**, access control, deadline validation, secure fund management, and checked arithmetic; implemented core market and betting instructions.
  Developed the **Next.js/React frontend** with TypeScript, wallet integration, and Codama-generated clients; added **LiteSVM tests** and security mechanisms including double-claim prevention and overflow protection.. *Aug 2026*

- **Open Source Project Analyzer [Live Demo] [GitHub]**:
  Tools & Technologies: **Rust**, **Next.js**, **TypeScript**, **PostgreSQL**, **SQLx**, **Docker**, **REST API**, **Google Gemini API**, **GitHub API**.
  Built a full-stack **Rust backend** with a **Next.js/TypeScript frontend** to analyze GitHub repositories using code-quality metrics and **Gemini AI**, with PostgreSQL persistence via **SQLx**.
  Implemented **REST APIs, GitHub API integration, database operations, AI analysis workflows**, and Docker-based deployment for automated repository quality assessment . *Aug 2026*

- **Decentralized Escrow System on Solana [GitHub]**:
  Tools & Technologies: **Rust**, **Solana**, **Anchor**, **SPL Token**, **Token-2022**, **LiteSVM**, **PDA**, **CPI**.
  Developed a **Solana smart contract** in **Rust/Anchor** for trustless peer-to-peer token swaps using **PDA-based custodial vaults**, atomic transactions, refunds, and on-chain state management.
  Implemented **CPI-based token transfers** with `transfer_checked`, SPL Token/Token-2022 interoperability, Anchor account validation, mint verification, secure account closure, and **LiteSVM testing**. *June 2026*

- **Solana On-Chain Voting Protocol [GitHub]**:
  Tools & Technologies: **Solana**, **Rust**, **Anchor**, **TypeScript**, **PDAs**, **LiteSVM**.
  Built an **on-chain voting smart contract** using **Rust/Anchor**, implementing poll creation, candidate registration, vote casting, state management, and **time-based access control** with PDA-based accounts.
  Developed **LiteSVM integration tests** covering account initialization, transactions, state updates, error handling, and voting-period validation. *May 2026*